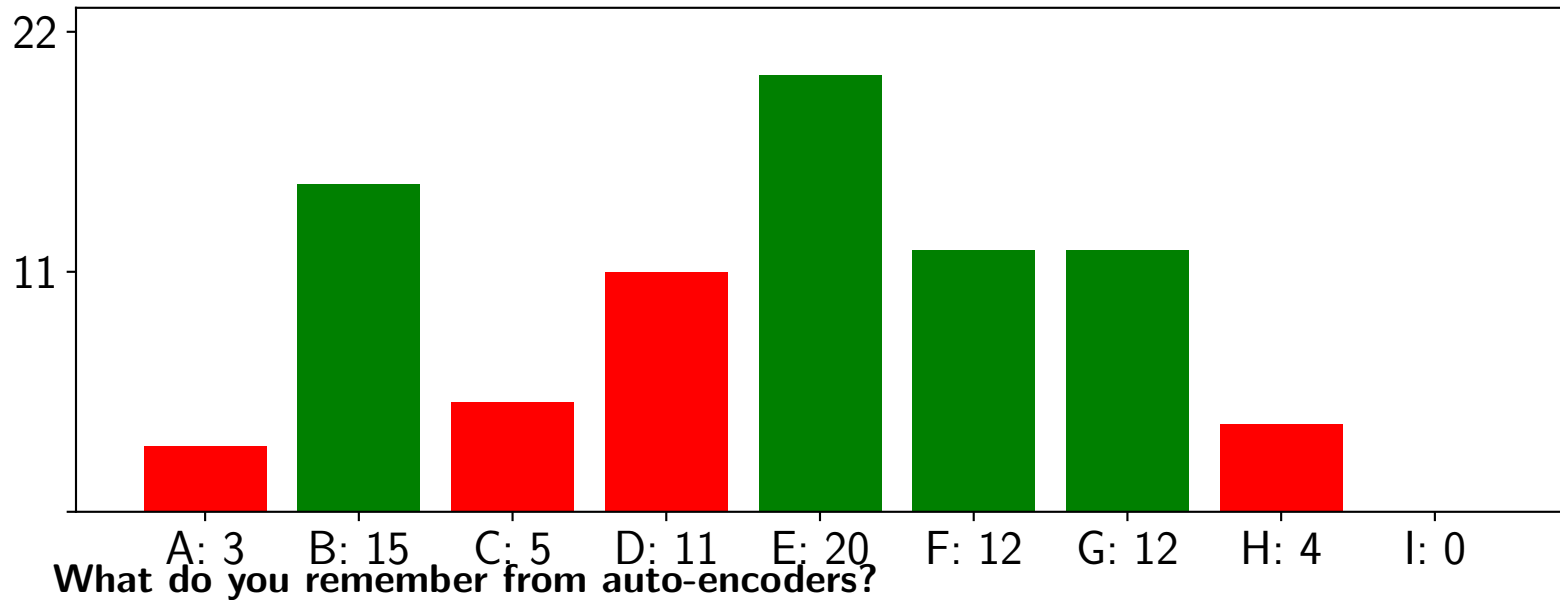
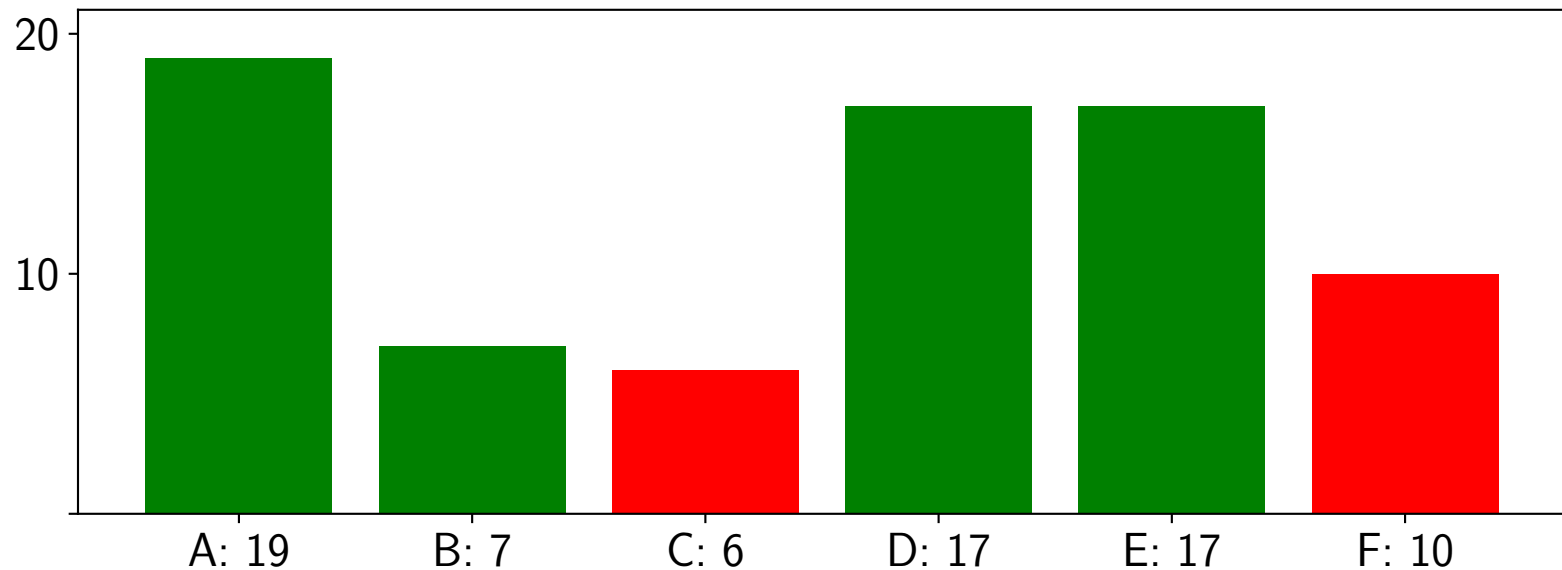


Auto-Encoders



- A Auto-encoders are a supervised machine learning technique.
- B The goal of auto-encoders is to learn a lower-dimensional representation of the input.
- C Auto-encoders are split into two parts: the generator and the discriminator.
- D A two-layer auto-encoder can learn a non-linear embedding of the features.
- E In a convolutional auto-encoder, transposed convolutions are used to reconstruct the input.
- F Fractionally-strided convolution is a valid option to increase the output of the convolution.
- G Variational auto-encoders regularize the feature space to be smooth.
- H Variational auto-encoders are typically used for feature extraction.
- I Diffusion models transform noise into data samples in one step.

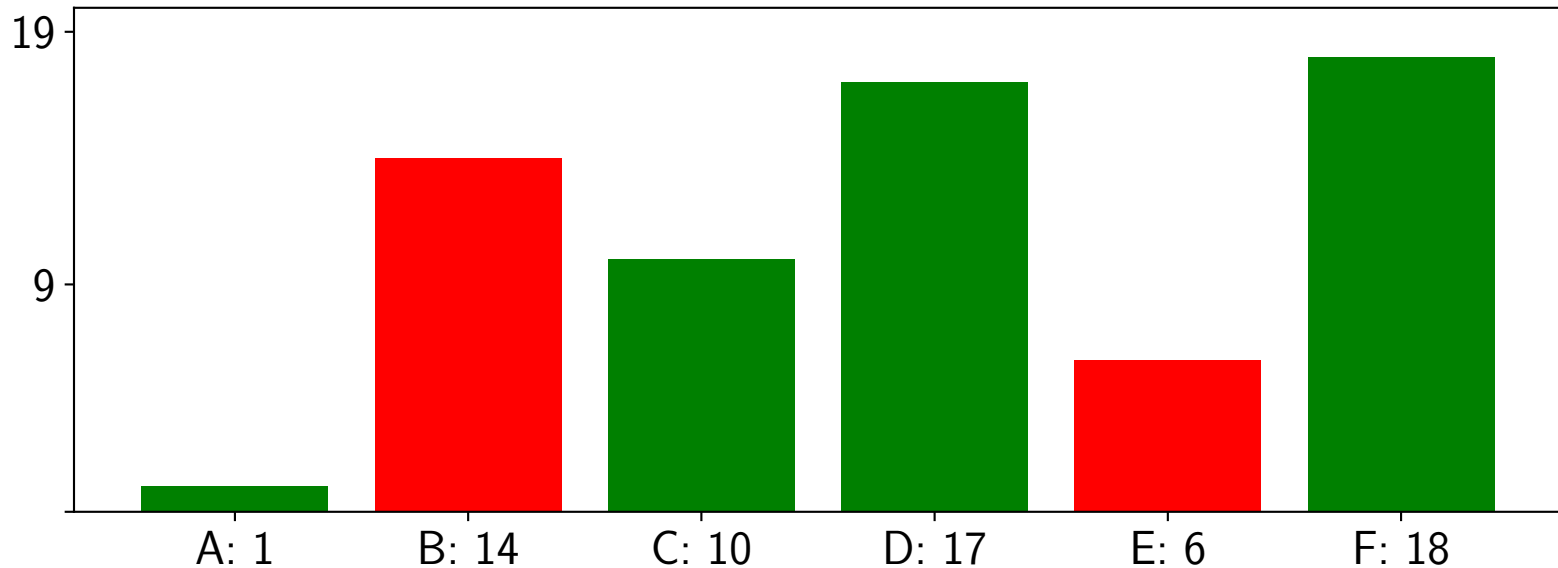
Generative Adversarial Networks



Here are some statements about generative adversarial networks.

- A Generative adversarial networks are split into two parts: a generator and a discriminator.
- B The goal of a generative adversarial network is to learn input distributions.
- C The generator reconstructs output data from the input data.
- D The discriminator tries to distinguish real from generated data.
- E The generator and the discriminator have competing goals.
- F The generator is trained to minimize the GAN loss.

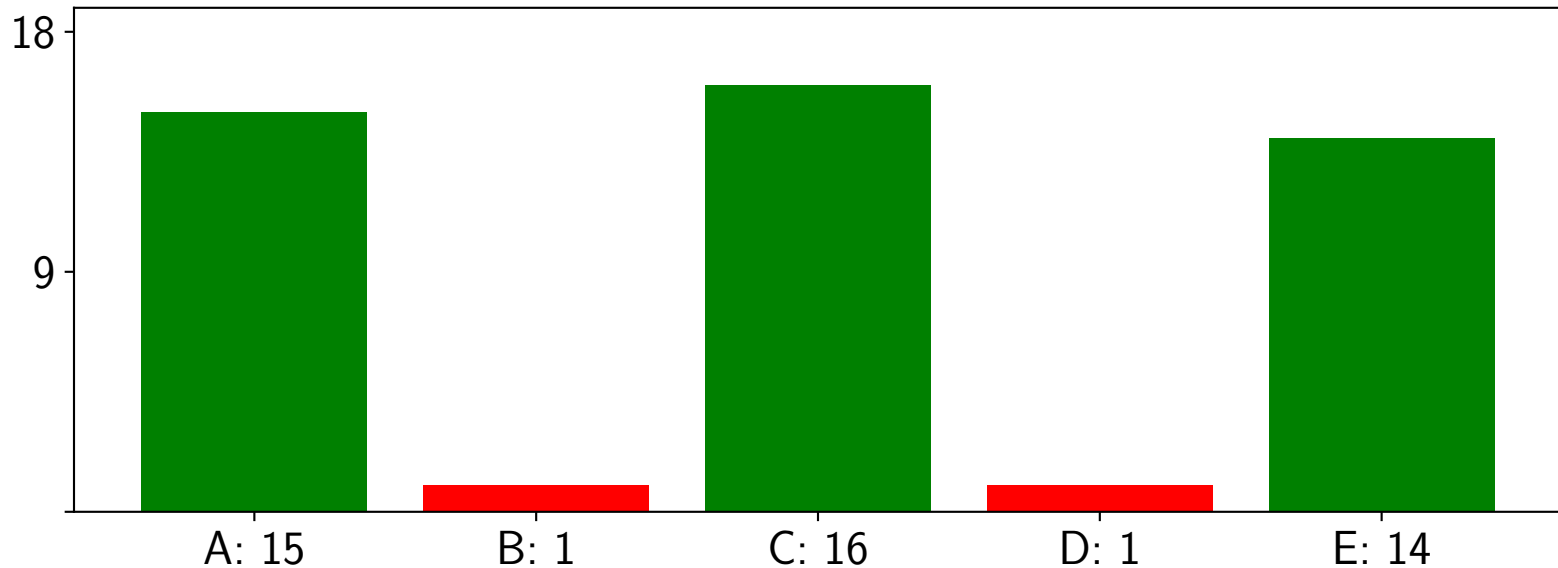
Recurrent Networks



Which statements on recurrent networks are correct?

- A All recurrent networks that we have seen in the lecture are feed forward networks.
- B The hidden representation $\vec{h}$ of the previous batch element is added to the input of the current batch element.
- C The output $\vec{h}^{\{S\}}$ of the last element can be used to represent the entire sequence.
- D Recurrent networks are trained via Backpropagation Through Time.
- E Recurrent networks are mainly applied to process textual information.
- F Simple recurrent networks have problems learning long-term relations between inputs.

Gated Recurrent Networks



What do you remember about gating mechanisms in recurrent networks?

- A Gating mechanisms allow information to be forwarded selectively.
- B Long-Short Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks are the only gated recurrent networks.
- C LSTM blocks include 3 gates: input, forget and output gate.
- D LSTM blocks include 6 learnable weight matrices, two per gate.
- E Bi-directional LSTM models combine two hidden representations per sequence element.